

SeismoFNO — Physics-Informed Neural Operators for Seismic Response

Raghvendra Singh Gahlot • B.E. Building & Construction Technology (Structural Engineering)
IIT Delhi CSE Research Internship Application • Computational Structural Mechanics

RESEARCH QUESTION:

Can topology-native and physics/modal-conditioned neural operators improve seismic structural-response prediction and generalization beyond the training structural distribution?

Structure

[M], [K]

Modal Anal.

Ground Mot.

Graph Rep.

Modal GNO

Response $u(t)$

KEY SCIENTIFIC PROGRESSION (EXP4 → EXP5 → EXP6)

- EXP4 — Fixed-Grid FNO:** 3S Rel L_2 = 99.60%. Failure: spatial zero-padding causes artificial boundary step & Gibbs ringing.
- EXP5 — Graph Neural Operator:** 3S Rel L_2 = 22.09% (77.51 pp drop). Topology limitation resolved via native floor nodes & column edges.
- EXP5 OOD-B Breakdown:** Unseen flexible frame 5S_T120 ($T_1=1.20s$ vs. train $T_1 \leq 0.85s$). Peak error = 35.21% (Modal extrapolation failure).
- EXP6-C — Multi-Modal GNO:** Peak error drops to 13.06% (62.9% relative error reduction via physical eigenvalue FiLM conditioning).
- EXP6-D — Shuffled Modal Control:** Peak error regresses to 24.33% (+11.27 pp), proving physical modal correspondence rather than auxiliary capacity.

EXP5 BASELINE

35.21%

Unconditioned OOD-B Error

EXP6-C MULTI-MODAL

13.06%

62.9% Relative Error Reduction

EXP6-D SHUFFLED

24.33%

Ablation Falsification Evidence

Model Architecture	In-Distribution (ID)	OOD-A (Seismic OOD)	OOD-B (Flexible 5S_T120)	OOD-C (Dual OOD)
EXP5 Baseline GNO	12.70%	15.86%	35.21%	38.66%
EXP6-B T1-GNO	2.09%	8.81%	13.47%	14.36%
EXP6-C Multi-Modal GNO	8.87%	7.63%	13.06%	17.01%

IMPORTANT LIMITATION: WAVEFORM PHASE DRIFT UNDER EXTRAPOLATION

Trajectory-level waveform fidelity remains poor under strong modal extrapolation (Rel $L_2 > 100\%$, Pearson $r \approx 0.05-0.09$). Global static 1D Fourier layers accumulate temporal phase discrepancy ($\Delta\theta = \Delta\omega \cdot t$) over 20.48 s. Peak response envelopes are accurately bounded, but continuous waveform tracking remains an open limitation.

MEASURED COMPUTATIONAL PERFORMANCE:

- OpenSeesPy C++ NLTHA: **54.68 ms** (18.3 sim/s)
 - EXP6 T1-GNO Surrogate: **21.45 ms** (46.6 sim/s) • **2.55x Speedup**
 - Batch Mode (B=32): **1,060 sim/s** (57.9x Throughput)
- Hardware: Apple Silicon GPU (MPS), Single-Building Transient Task

REPRODUCIBILITY & SCIENTIFIC FREEZE:

- Test Suite: **294 Passed, 2 Skipped, 0 Failed**
 - Forensic Audit: **5/5 Verification Checks Passed**
 - Zero retraining, dataset edits, or metric modifications.
- 2,160 Physical Simulations • Zero Partition Leakage

FINAL THESIS:

"Graph-native neural operators remove the fixed-grid topology limitation encountered by conventional FNO representations, while physics-informed modal conditioning improves peak-response generalization under structural modal-period shift. Trajectory-level waveform fidelity remains an open limitation."